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6.042J

Mathematics for Computer Science 6.042J — Mathematics for Computer Science May

2026 **EXTREMELY HARD – MIT LEVEL****Time Limit: 3 hours****Total Marks: 100**

1. (4 points) Prove that one-way functions exist if and only if $P \neq NP$. Construct a one-way function based on the hardness of SAT, or show that the existence of OWFs is equivalent to pseudorandom generators.
2. (2 points) Prove type soundness for the simply typed lambda calculus using the progress and preservation lemmas. Extend the proof to System F with parametric polymorphism.
3. (5 points) Prove the Goldreich-Levin theorem: if f is a one-way function, then there exists a one-way function g and a hardcore predicate b such that $b(x)$ is hard to predict given $g(x)$.
4. (5 points) Prove that the amortized cost of a sequence of n operations in a Fibonacci heap is $O(\log n)$ for extract-min and $O(1)$ for all other operations, using the potential function method.
5. (4 points) Prove the CAP theorem: a distributed data store cannot simultaneously provide more than two of Consistency, Availability, and Partition tolerance. Formalize the theorem using a proof by contradiction with atomic registers.
6. (5 points) Prove that parity cannot be computed by a polynomial-size constant-depth circuit (AC^0) using the switching lemma. Show that any AC^0 circuit computing parity must have size $\exp(\Omega(n^{1/(d-1)}))$.
7. (5 points) Prove that the matrix multiplication exponent ω satisfies $2 \leq \omega < 2.373$ by analyzing the Coppersmith-Winograd algorithm and its subsequent improvements.
8. (2 points) Prove the Byzantine Generals Problem has no solution for $n \leq 3m$ generals with m traitors in the oral messages model. Show the $n \geq 3m + 1$ necessary condition.
9. (3 points) Prove an $\Omega(n \log n)$ lower bound for sorting in the comparison-based model using the adversary method. Then apply the adversary argument to prove a lower bound for finding the k -th smallest element.

10. (4 points) Prove the Karger-Stein randomized min-cut algorithm runs in $O(n^2 \log^3 n)$ time and succeeds with high probability. Analyze the probability that a particular min-cut survives the contraction process.
11. (2 points) Prove the VC-dimension bound for PAC learning: if a hypothesis class $\mathcal{H}$ has VC-dimension d , then the sample complexity for (ϵ, δ) -PAC learning is $m \geq \Omega(\frac{1}{\epsilon}(d + \log(1/\delta)))$. Provide the upper bound as well.
12. (7 points) Prove that problem $\text{RL} \subseteq \text{SC}$ (randomized logspace is in Steve's class) using the Nisan generator and showing that RL can be derandomized with polynomial seed length.
13. (4 points) Prove that graph isomorphism is in $\text{NP} \cap \text{coAM}$. Show that GI is unlikely to be NP-complete (if it were, PH collapses).
14. (5 points) Prove that 3-SAT reduces to Hamiltonian Cycle in polynomial time. Construct the explicit gadget for each clause and variable, showing that a satisfying assignment corresponds to a Hamiltonian cycle in the constructed graph.
15. (2 points) Prove that the minimum distance of a linear code can be computed using the parity check matrix and show that the decisional version is NP-complete (Berlekamp-Welch theorem variant).
16. (5 points) Prove the PCP theorem: $\text{NP} = \text{PCP}(\log n, 1)$. Construct a verifier for SAT that reads $O(1)$ bits of the proof and uses $O(\log n)$ random bits.
17. (6 points) Prove that if $\text{P} = \text{NP}$, then $\text{BPP} = \text{P}$. Conversely, show that $\text{BPP} \subseteq \text{EXP}$ and discuss the relationship between P and BPP.
18. (3 points) State and prove Rice's theorem: every nontrivial semantic property of partial recursive functions is undecidable. Show that 'program P halts on input 0' is such a property.
19. (3 points) Prove that EXP^{NP} contains languages that require exponential-size circuits (Karp-Lipton theorem variant). Show that if $\text{NP} \subseteq \text{P/poly}$ then $\text{PH} = \Sigma_2^P$.
20. (4 points) Prove correctness of the Cheney garbage collection algorithm using a formal argument about the copy phase preserving the reachability graph.
21. (3 points) Prove the Adleman theorem: $\text{BPP} \subseteq \text{P/poly}$. Show that any randomized polynomial-time algorithm can be derandomized with polynomial-size advice.
22. (4 points) Prove the parallel repetition theorem for interactive proofs: if a verifier accepts with probability p in a single round, then the verifier accepts in n parallel repetitions with probability at most $p^{\Omega(n)}$ (for certain protocols).

23. (2 points) Prove the Cook-Levin theorem: SAT is NP-complete. Construct a polynomial-time reduction from any NP language L to SAT.
24. (7 points) Prove the space hierarchy theorem: for space-constructible $f(n) = \Omega(\log n)$, $\text{SPACE}[f(n)] \subsetneq \text{SPACE}[f(n) \log f(n)]$.
25. (4 points) Prove the Chernoff bound: for independent Bernoulli random variables $X_1, \dots, X_n$ with $\Pr[X_i = 1] = p_i$, let $X = \sum X_i$ and $\mu = \mathbb{E}[X]$. Show that $\Pr[X \geq (1 + \delta)\mu] \leq (\frac{e^\delta}{(1+\delta)^{1+\delta}})^\mu$ for all $\delta > 0$.

Solutions

Solution 1:

($\Rightarrow$): If OWF f exists, then inverting f is in NP (guess a preimage and verify) but not in P by definition. So $P \neq NP$. ($\Leftarrow$): Unconditionally constructing OWF from $P \neq NP$ is open. However, using Levin's construction, we can define $f(M, x, 1^t) = (M, x, 1^t)$ where M outputs the result of t steps, which is one-way if $P \neq NP$ under certain assumptions.

Solution 2:

Progress: if $\vdash e : \tau$ and e is closed, then e is a value or $e \rightarrow e'$. Proof by induction on typing derivation. Preservation: if $\vdash e : \tau$ and $e \rightarrow e'$, then $\vdash e' : \tau$. Proof by induction on the reduction relation with substitution lemma. For System F, add type abstraction and application cases, with type substitution preserving the subject reduction property.

Solution 3:

Define $g(x, r) = (f(x), r)$ for $|x| = |r|$. The hardcore predicate $b(x, r) = x \cdot r \bmod 2$. If an adversary predicts b with advantage ε , construct an inverter for f : use a clever reconstruction (Goldreich-Levin list decoding) that, given $f(x)$ and oracle access to the predictor, outputs a list containing x with high probability. The reconstruction uses pairwise independent sampling and majority decoding.

Solution 4:

Define $\Phi = \text{number of trees} + 2 \cdot \text{number of marked nodes}$. Insert: add one tree, $\Delta\Phi = 1$, actual cost $O(1)$, amortized $O(1)$. Decrease-key: cut and cascade, each cut adds a tree but may clear marks. Extract-min: $O(\text{trees} + \text{degree})$ actual cost. By the rank bound $\text{degree} = O(\log n)$, and trees decrease by at most $O(\log n)$, giving amortized $O(\log n)$.

Solution 5:

Assume all three hold. During a partition, two nodes cannot communicate. A write to node A must be visible on a read from node B (consistency). But node B must respond (availability) without contacting A (partition). So B has either stale data (no consistency) or must wait (no availability), contradiction. This shows at most two properties can hold simultaneously.

Solution 6:

Håstad's switching lemma: a random restriction of a DNF formula leaves a decision tree of depth $O(\log s)$ with high probability. Apply this to each layer of an AC^0 circuit of depth d and size S . After $d - 1$ random restrictions, the circuit becomes a decision tree of depth $O(\log S)^{d-1}$. Since parity has decision tree depth n , $O(\log S)^{d-1} \geq n$, so $S \geq 2^{\Omega(n^{1/(d-1)})}$.

Solution 7:

Strassen's $O(n^{2.807})$ algorithm uses the identity $M_2(2) = 7$ giving $\omega = \log_2 7$. Coppersmith-Winograd uses the laser method and bilinear algebra to achieve $\omega < 2.376$. The key idea: construct a family of tensors T_n with border rank r_n and combining them recursively. The asymptotic inequality $\omega < \log_{n^{2/3}}(r_n)$ yields improved bounds. The current best $\omega < 2.37286$ uses refined laser method analysis.

Solution 8:

For $m = 1$, $n = 3$: assume generals A (commander), B, C with B loyal, C traitor. Commander sends 'attack'. C tells B 'retreat'. B receives conflicting messages. For $n = 3m$, reduce by grouping: simulate each general as $m + 1$ generals, one traitor in each group to create inconsistency. The $n \geq 3m + 1$ condition is sufficient via the Oral Message algorithm OM(m) recursively sending $m + 1$ rounds.

Solution 9:

Any comparison tree that sorts n elements must have at least $n!$ leaves, so depth $\geq \log_2(n!) = \Omega(n \log n)$. Adversary argument: maintain a set of possible permutations consistent with comparisons so far. Each comparison eliminates at most half the remaining permutations, so $\Omega(\log(n!))$ comparisons needed. For selection, the adversary maintains a partial order ensuring at least $n - 1$ comparisons are needed for the minimum.

Solution 10:

Contract edges uniformly until t vertices remain. A fixed min-cut C of size k survives a single contraction step with probability $\geq 1 - 2/t$. By induction, survival probability $\geq \binom{t}{2}/\binom{n}{2}$. Setting $t = n/\sqrt{2}$ and recursing twice gives $O(n^2 \log n)$ expected time. The success probability is $\Omega(1/\log n)$, and repeating $O(\log^2 n)$ times gives high probability.

Solution 11:

Lower bound: construct d shattered points. The adversary picks a target concept uniformly from 2^d dichotomies. To learn, we must see each point with label to distinguish. By the Sauer-Shelah lemma, $|\mathcal{H}(m)| \leq (em/d)^d$. Setting sample size $m = O(\frac{1}{\epsilon}(d \log(1/\epsilon) + \log(1/\delta)))$ gives the upper bound via the ERM principle.

Solution 12:

Nisan's pseudorandom generator for logspace: $G : \{0, 1\}^{O(\log^2 n)} \rightarrow \{0, 1\}^n$ fools any logspace machine. The construction uses the INW generator (impagliazzo-Nisan-Wigderson) based on extractors. By enumerating all $2^{O(\log^2 n)}$ seeds, we can simulate RL in SC (polynomial time, polylog space). This gives $\text{RL} \subseteq \text{SC} \subseteq \text{P}$.

Solution 13:

$\text{GI} \in \text{NP}$: a permutation is a polynomial-time verifiable certificate. $\text{GI} \in \text{coAM}$: the verifier sends a random permutation of one graph; the prover must identify which graph. The interactive protocol has soundness $1/2$. If GI were NP-complete, then $\text{PH} = \Sigma_2^P$ (Boole'an-Karp'n theorem) because $\text{coNP} \subseteq \text{AM}$ and AM collapses if $\text{coNP} \subseteq \text{AM}$.

Solution 14:

For each variable x_i , create a chain of vertices $v_{i,1}, \dots, v_{i,k}$ with edges forming a path that can be traversed left-to-right (true) or right-to-left (false). For each clause $C_j = (\ell_1 \vee \ell_2 \vee \ell_3)$, insert a clause vertex connected to the corresponding literal vertices in the variable chains. A Hamiltonian cycle exists iff we can pick a consistent direction for each variable chain that hits all clause vertices. This is the standard NP-completeness reduction.

Solution 15:

For a linear $[n, k]$ code with parity check matrix H , the minimum distance $d =$

$\min_{c \neq 0, Hc^T=0} w(c)$. Decisional problem (given G and w , is $d \leq w$?) is NP-complete via reduction from 3SAT, by constructing a code whose parity checks encode clause satisfaction. For a linear code over $\mathbb{F}_q$, finding minimum weight codeword is NP-hard.

Solution 16:

$(\supseteq)$ is trivial. $(\subseteq)$: Build a PCP verifier for SAT. Start from the NP-complete constraint satisfaction problem with constant degree (via expander graphs). Apply the gap amplification step (repeatedly replace constraints by their AND using composition). The final verifier reads $O(1)$ proof bits. The theorem shows that approximate solutions to NP problems remain hard.

Solution 17:

If $P = NP$, then $PH = P$ (polynomial hierarchy collapses). Since $BPP \subseteq \Sigma_2^P \cap \Pi_2^P$ (by Sipser-Gacs theorem), we get $BPP \subseteq P$. The inclusion $BPP \subseteq EXP$ follows from exhaustive simulation of the probabilistic computation over all random bits. Whether $P = BPP$ is open but believed true due to pseudorandom generators.

Solution 18:

Let $\mathcal{P}$ be a nontrivial property of partial recursive functions, i.e., there exist f, g with $f \in \mathcal{P}$, $g \notin \mathcal{P}$. Assume $\mathcal{P}$ is decidable. Reduce the halting problem to it: for input (M, x) , construct M' that on input y simulates M on x and then behaves like f (if $\emptyset \notin \mathcal{P}$) or like g (if $\emptyset \in \mathcal{P}$). Then M halts on x iff $M' \in \mathcal{P}$, a contradiction.

Solution 19:

If $NP \subseteq P/\text{poly}$, then SAT has polynomial-size circuits. By the Karp-Lipton theorem, $\Pi_2^P \subseteq \Sigma_2^P$, collapsing PH to Σ_2^P . The proof: for any Π_2^P language, the existential player guesses the circuit for SAT and uses it to answer the universal quantifier. This shows $PH \subseteq \Sigma_2^P$, hence $PH = \Sigma_2^P$.

Solution 20:

Cheney's algorithm uses a semi-space collector. Maintain *fromspace* and *tospace*. The scan pointer iterates through *tospace* while the alloc pointer advances. Invariant: all reachable objects from roots have been copied or are in the queue. Each edge traversed preserves connectivity: $\forall x$ reachable from root, x is either in *tospace* (scanned or unscanned) or reachable from unscanned objects. Termination: when scan = free, all reachable objects are in *tospace*.

Solution 21:

Let $L \in BPP$ with TM M using $m(n)$ random bits. For each input size n , the set of bad random strings for input x has size $\leq 2^{m(n)}/3$. By union bound over 2^n inputs, the total bad set has size $\leq 2^n \cdot 2^{m(n)}/3 < 2^{m(n)}$ for n large enough. So there exists a good random string r_n that works for all inputs of length n . Provide r_n as advice, giving $BPP \subseteq P/\text{poly}$.

Solution 22:

For XOR games, Raz's parallel repetition theorem shows the value decays exponentially: $\omega(G^{\otimes n}) \leq \omega(G)^{cn}$ where c depends on the game. The proof uses information theory: the winning probability for each coordinate is correlated only through the ini-

tial shared randomness, and the protocol cannot coordinate between coordinates without increasing communication.

Solution 23:

Let $L \in \text{NP}$ via NTM M running in time $p(n)$. Construct a boolean formula ϕ with variables encoding the computation tableau of M on input x of size n . ϕ has clauses for: (1) initial configuration, (2) valid transitions using $O(p(n)^3)$ variables, (3) accepting state. ϕ is satisfiable iff M accepts x . The reduction runs in polynomial time, proving SAT is NP-hard.

Solution 24:

Diagonalization: construct a language L decidable in $O(f(n) \log f(n))$ space but not in $o(f(n))$ space. The diagonalizing machine D on input x simulates M_x on x using at most $f(|x|)$ space and if M_x halts within $2^{O(f)}$ steps, outputs the opposite. The simulation uses a counter of $O(\log f)$ bits. The $\log f(n)$ factor is needed for configuration counting to detect infinite loops.

Solution 25:

Use the moment generating function: $\Pr[X \geq t] \leq \min_{\lambda > 0} e^{-\lambda t} \mathbb{E}[e^{\lambda X}] = \min_{\lambda > 0} e^{-\lambda t} \prod \mathbb{E}[e^{\lambda X_i}]$. Since $\mathbb{E}[e^{\lambda X_i}] = 1 + p_i(e^\lambda - 1) \leq \exp(p_i(e^\lambda - 1))$, we get $\Pr[X \geq (1 + \delta)\mu] \leq (\frac{e^\delta}{(1+\delta)^{1+\delta}})^\mu$ by optimizing $\lambda = \ln(1 + \delta)$.